MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC
GATE, NAND, DIGITAL, KEY CODE BDC

INACTIVE FOR NEW DESIGN
AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module

1 2 Classification This specification covers one class of modules

Class I as specified in MIL-STD-1389.

2 APPLICABLE DOCUMENTS

2 1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3 REQUIREMENTS

3 1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3 2 Correlation requirements Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

3 3 Failure rate requirements All modules acquired to this specification shall have a predicted failure rate no greater than 2 10 failures per million operating hours when predicted in accordance with MIL-STD-1389

3 4 Performance requirements The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389

3 5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3 5 1 Key code The module key code shall be BDC

3 5.2 Size The module size shall be 1A

3 5.3 Weight The module shall have a maximum weight of 75 grams

3 5 4 Contact pins The module shall have 40 contact pins positioned A1 through A40

3 5.5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth. The module shall have no additional depth requirements

3.5.7 Holddown holes The module shall have no additional requirements for location or quantity of holddown holes

3 5.8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/116-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3 6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1

3 7 Acceptance tests All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3 8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6 2)

3 9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5 0 ±0 5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5 5 Vdc ±1 percent

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

4 2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4 3 Test equipment Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

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4.4 Test procedures The module shall be tested as specified in table I

4.4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787

4.4.2 Order of testing Test conditions A through R shall be performed in any order with the following exceptions If test conditions A, B, C, or D1 are part of a sequence of tests, the order of testing shall be as follows A, B, C, D1 where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4.5.1 Operating temperature tests The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions E through J and L through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2

4.5.1.1 Component temperature test The component temperature test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 3

4.5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 4 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests E through J and L through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests E through R shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I

5 PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

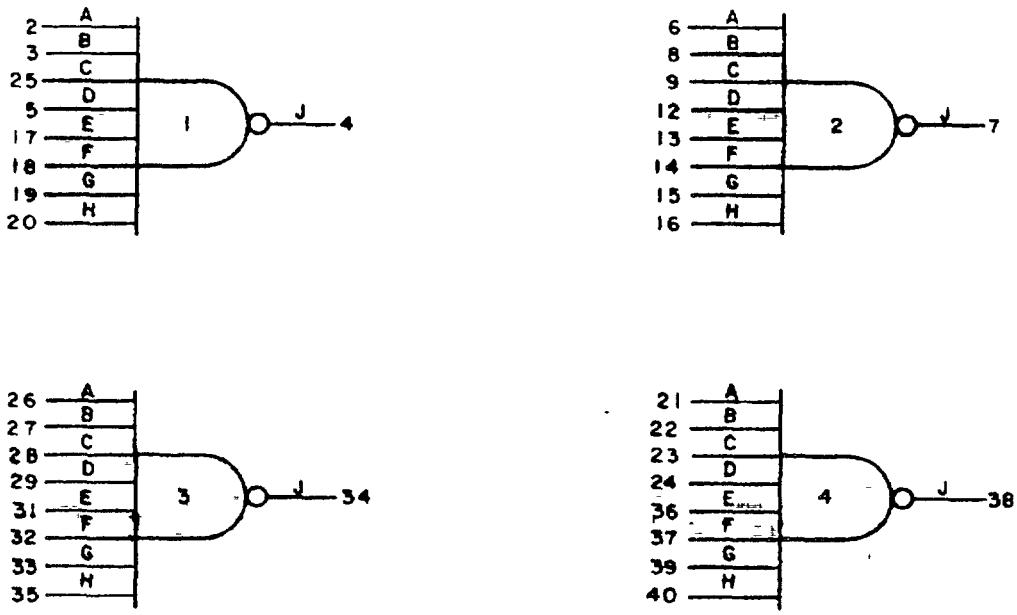
6 NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8)

6.3 Cross reference of classification Military specification M28787/116 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2658601 dated 3 September 1971 Part number M28787/116-1 supersedes part numbers 2658636 and M28787/116 These parts are interchangeable

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NOTES
1. Dashed line denotes optional connection internal to module
2. A through H = Gate inputs
3. J = Gate output

FIGURE 1. BDC function diagram, NAND gate

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TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A32	Zener applied potential, condition A	III	S	3 8	5 5	3 5	5 5	3.8	5.5	3 5	5 5	V
B1-B4	Zener applied potential, condition B	III	S	0 00	0.40	0.00	0.44	0.00	0.42	0.00	0 46	V
C1-C4	V_{OUT} (low loaded)	IV	S	0 0	1 1	0 0	1.1	0 0	1 1	0 0	1 1	V
D1	Isolation	V	S	10	∞	10	∞	10	∞	10	∞	MΩ
D2	Continuity	V	S	0 0	1 0	0 0	1 0	0 0	1 0	0.0	1 0	Ω
E1-E32	Output voltage, high level	VI	X	3.8	5 5	3 5	5 5	3.8	5 5	3 5	5 5	V
F1-F4	Output voltage, low level	VI	X	0 00	0 40	0 00	0 44	0 00	0 42	0 00	0.46	V
G1-G4	Output short circuit current	VII	X	0 70	1 31	0 70	1.31	0 70	1 31	0 70	1.31	mA
H1-H32	Input current, low level	VIII	X	0 2	1 6	0 2	1 6	0 2	1 6	0 2	1.6	mA
J1-J224	Input current, high level	VIII	X	0	2	0	4	0	3	0	5	µA
K1-K32	Input capacitance	IX	Q	2	16	2	16	2	16	2	16	pF
L1	Power supply current, high level	X	S	0	11	0	11	0	11	0	11	mA
L2	Power supply current, low level	X	S	0 0	17 6	0 0	17.6	0 0	17 6	0 0	17 6	mA

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M	Power supply transient rise time	XI	S	36	148	36	148	36	148	36	148	µs
N1-N4	Output pulse amplitude, high level	XII	S	3 8	5 5	3 5	5 5	3 8	5 5	3 5	5 5	V
N5-N8	Output pulse amplitude, low level	XII	S	0 00	0 40	0 00	0 44	0 00	0 42	0 00	0 46	V
P1-P4	Transition time (t_{THL})	XIII	S	6	24	6	24	6	24	6	24	ns
P5-P8	Transition time (t_{TLH})	XIII	S	60	160	60	160	60	160	60	160	ns
P9-P12	Propagation delay time (t_{PLH})	XIII	S	20	75	20	75	20	75	20	75	ns
P13-P16	Propagation delay time (t_{PHL})	XIII	S	6	70	6	70	6	70	6	70	ns
Q1-Q4	Output crosstalk, high level	XIV	S	0 0	0 8	0 0	0 8	0 0	0 8	0 0	0 8	V
R1-R4	Output crosstalk, low level	XIV	S	0 0	0 8	0 0	0 8	0 0	0 8	0 0	0 8	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

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6.4 Module description. The following electrical parameters reflect the 0 to 60°C end-of-life limits of table I

Power dissipation	
All outputs at low level	97 mW maximum (outputs not loaded)
All outputs at high level	61 mW maximum (outputs not loaded)
Input voltage	
High level	1.8 V minimum, V_{CC} maximum
Low level	-0.5 V minimum, 1.2 V maximum
Input current	
High level	5.0 µA maximum
Low level	-1.6 mA maximum
Output voltage	
High level	3.5 V minimum, V_{CC} maximum
Low level	0.0 V minimum, 0.46 V maximum
Output drive current	
High level	-60 µA maximum
Low level	11.2 mA maximum
Propagation delay time	
t_{PLH}	75 ns maximum
t_{PHL}	70 ns maximum

6.5 Module function. The module consists of four 8-input NAND gates with independent inputs and outputs

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship for positive logic shown in table II

6.5.2 Paralleling of outputs. Paralleling of outputs is permissible with a reduction in low level fan out calculated from the following expression

$$\text{Reduced low level fan out} = \text{maximum fan out} - \left(\frac{5}{6}\right)(n - 1)$$

where maximum fan out = 7 SLL (Standard Logic Loads, Low level) and n = the number of outputs connected in parallel

The increase in high level fan out is calculated from the following expression

$$\text{Increased high level fan out} = (\text{Maximum fan out})(n)$$

where maximum fan out = 12 SLL (Standard Logic Loads, High level) and n = the number of outputs connected in parallel

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the precious issue, due to the extensiveness of the changes

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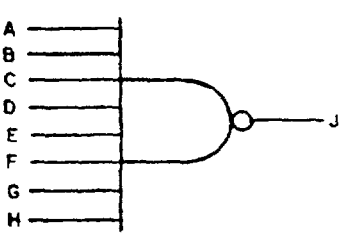
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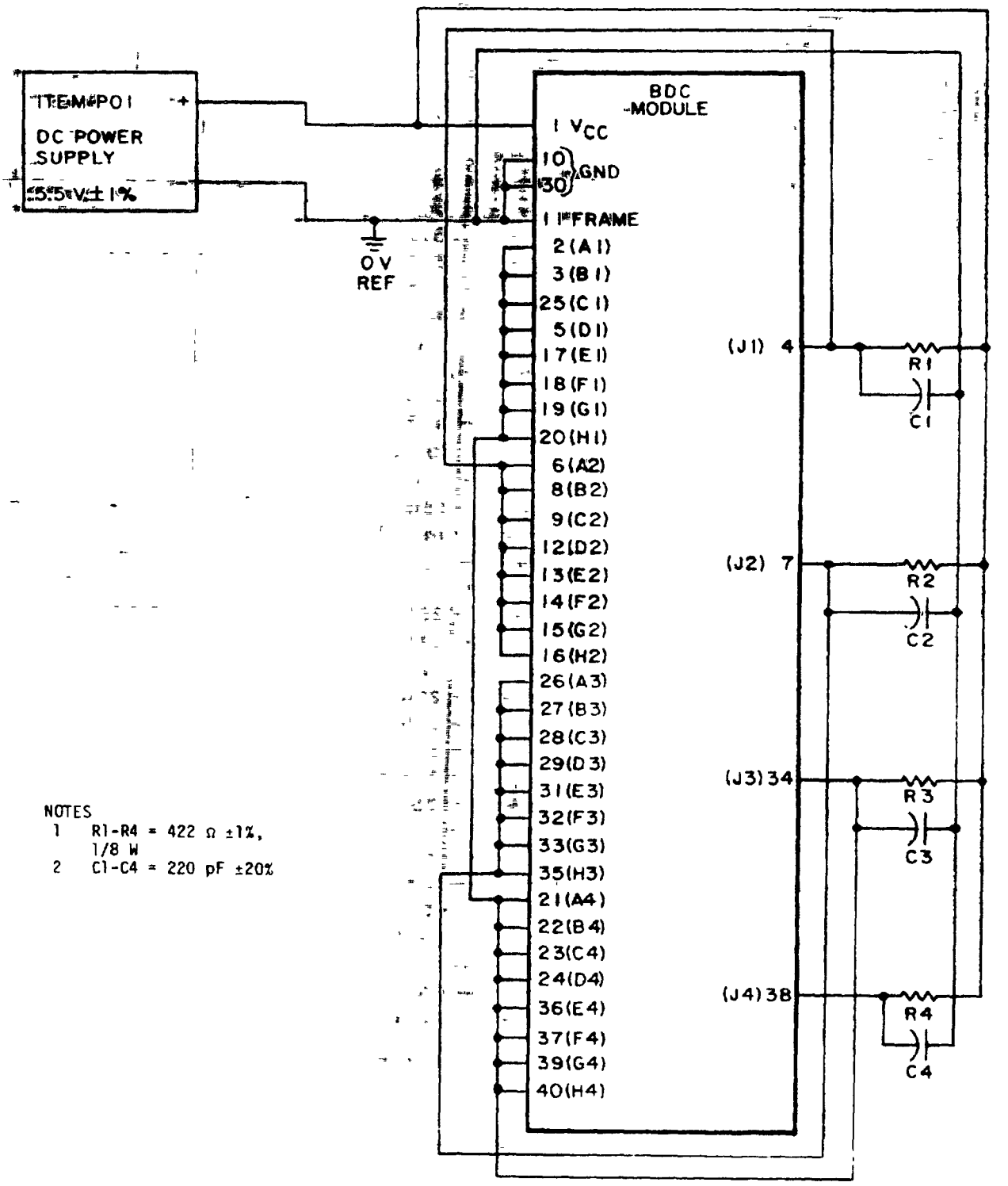
TABLE II Input to output relationship for each NAND gate

INPUT								OUTPUT
A	B	C	D	E	F	G	H	J
L	X	X	X	X	X	X	X	H
X	L	X	X	X	X	X	X	H
X	X	L	X	X	X	X	X	H
X	X	X	L	X	X	X	X	H
X	X	X	X	L	X	X	X	H
X	X	X	X	X	L	X	X	H
X	X	X	X	X	X	L	X	H
X	X	X	X	X	X	X	L	H
H	H	H	H	H	H	H	H	L



1/ A through H = Gate inputs
2/ J = Gate output
3/ H = High level voltage
4/ L = Low level voltage
5/ X = Don't care

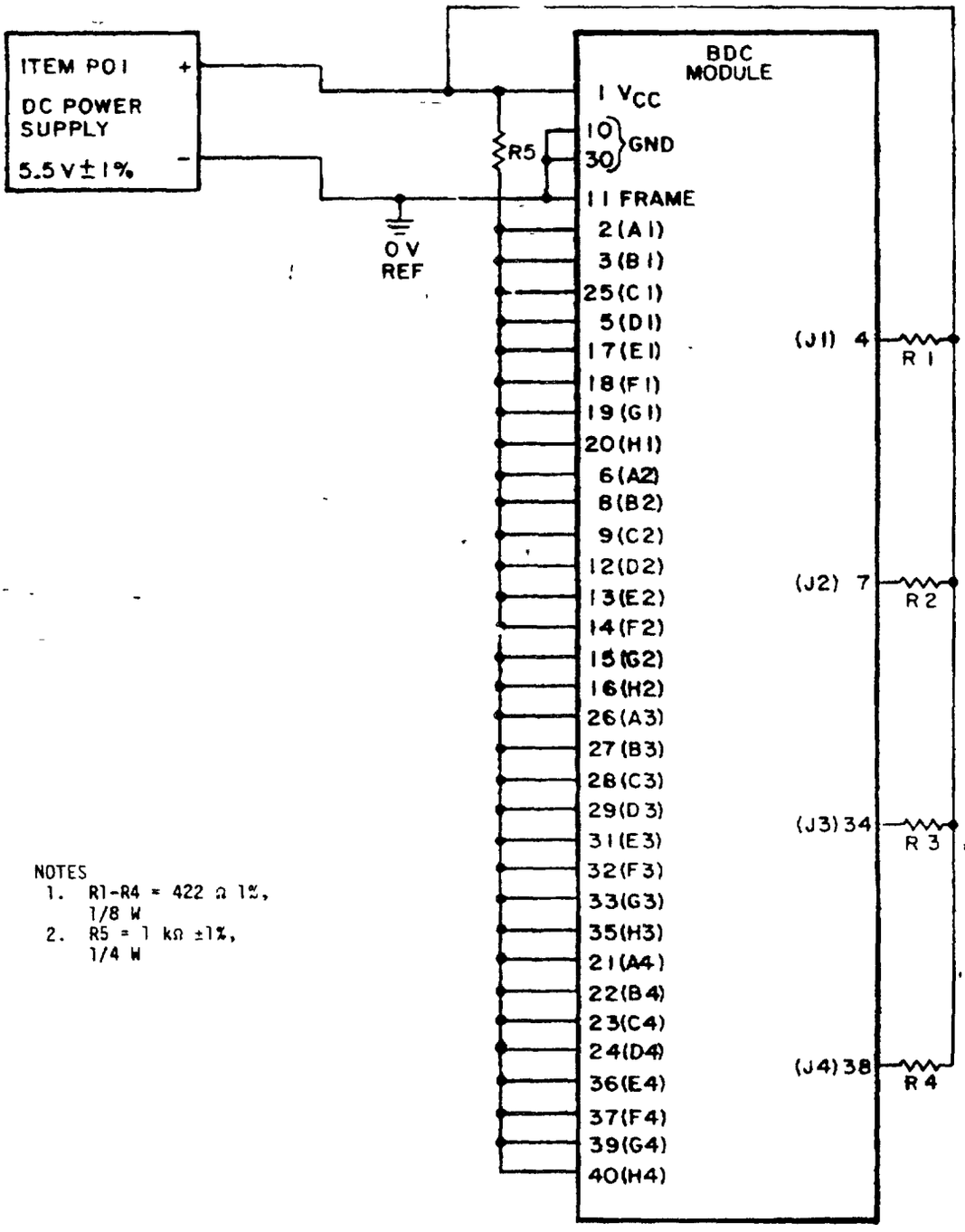
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NOTES
1. R1-R4 = 422 Ω ±1%, 1/8 W
2. C1-C4 = 220 pF ±20%

FIGURE 2 Operating temperature and life test circuit.

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NOTES
1. R1-R4 = 422 Ω 1%, 1/8 W
2. R5 = 1 kΩ ±1%, 1/4 W

FIGURE . Component temperature test circuit.

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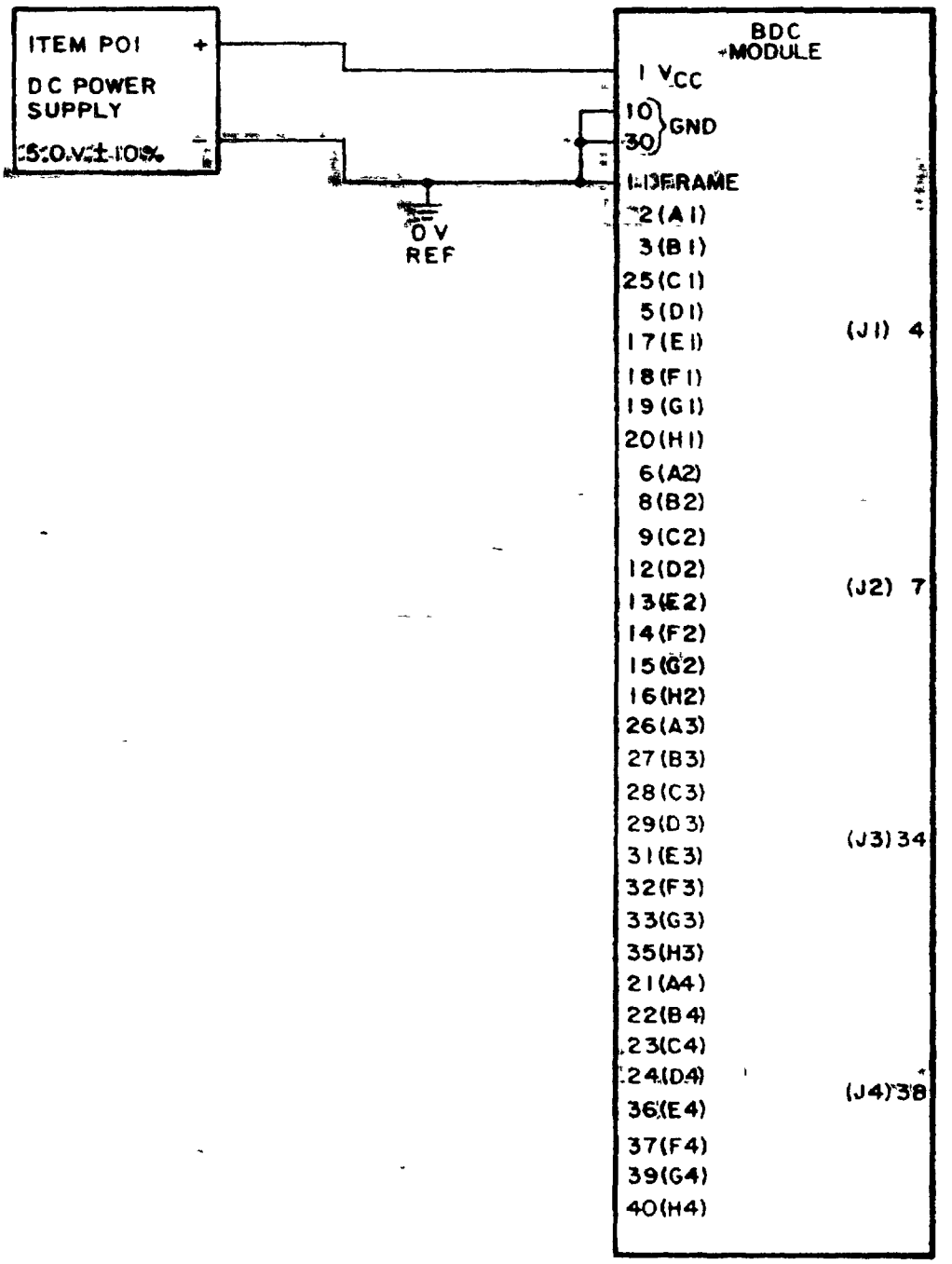


FIGURE 4 Vibration test circuit

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TABLE III Zener applied potential test

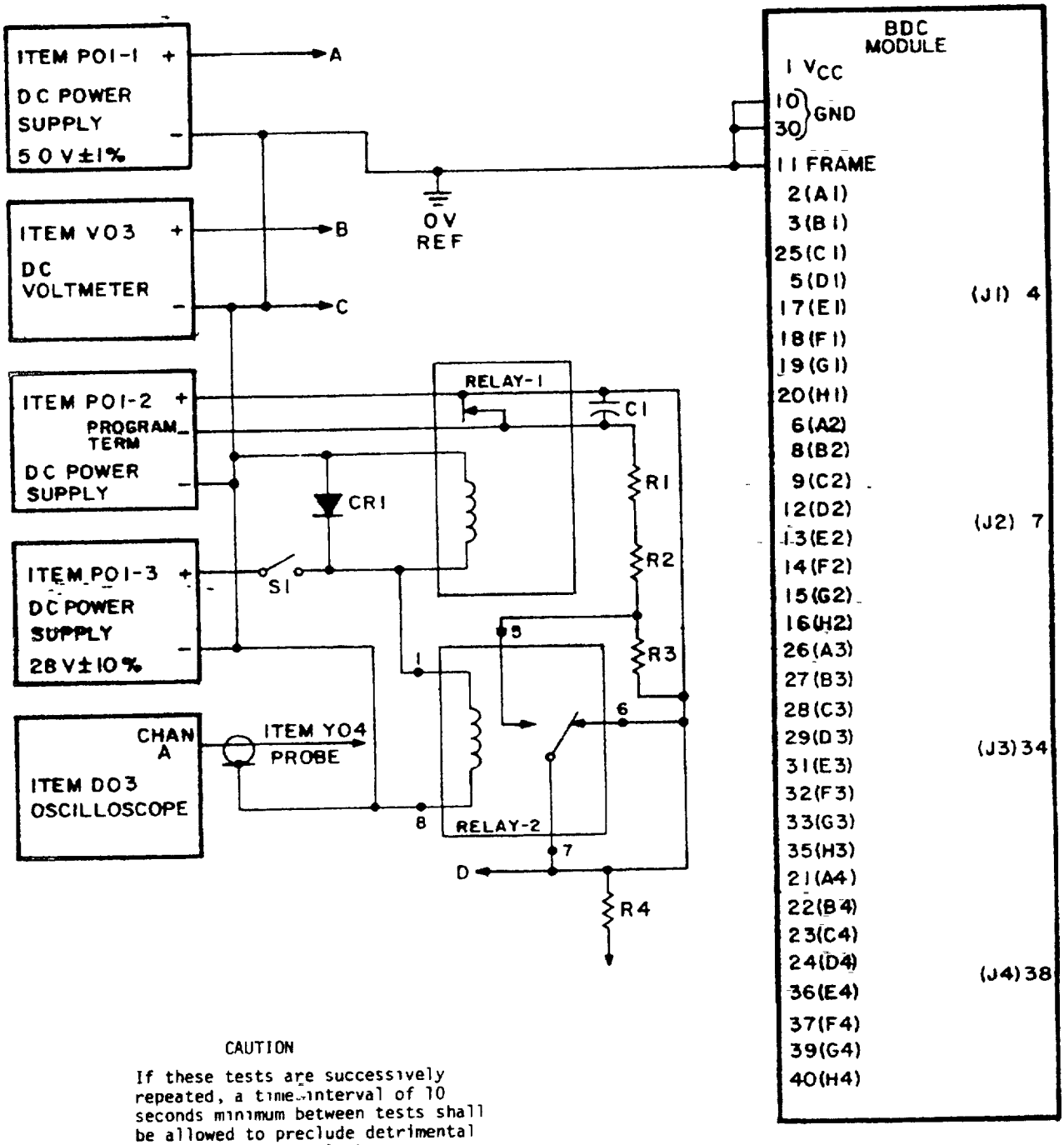
Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A1	J1 Zener applied potential, condition A	Open	4	3	1	2	3.8	5.5	V	5.6
A2	J1			2		3				
A3	J1			25		25				
A4	J1			25		5				
A5	J1			18		17				
A6	J1			17		18				
A7	J1			20		19				
A8	J1			19		20				
A9	J2		7	8		6				
A10	J2			6		8				
A11	J2			12		9				
A12	J2			9		12				
A13	J2			14		13				
A14	J2			13		14				
A15	J2			16		15				
A16	J2			15		16				
A17	J3		34	27		26				
A18	J3		34	26		27				

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TABLE III Zener applied potential tests - Continued

Test condition	Parameter	Connect					Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A19	J3 Zener applied potential, condition A	Open	34	29	1	28	3.8	5.5	V	5,6
A20	J3			28		29				
A21	J3			32		31				
A22	J3			31		32				
A23	J3			35		33				
A24	J3			33		35				
A25	J4		38	22		21				
A26	J4		38	21		22				
A27	J4			24		23				
A28	J4			23		24				
A29	J4			37		36				
A30	J4			36		37				
A31	J4			40		39				
A32	J4			39		40				
B1	J1 Zener applied potential, condition B	1	4	Open	Open	2	0.00	0.40		
B2	J2		7			6				
B3	J3		34			26				
B4	J4		38			21				

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CAUTION

If these tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

FIGURE 5 Zener applied potential test circuit for table III

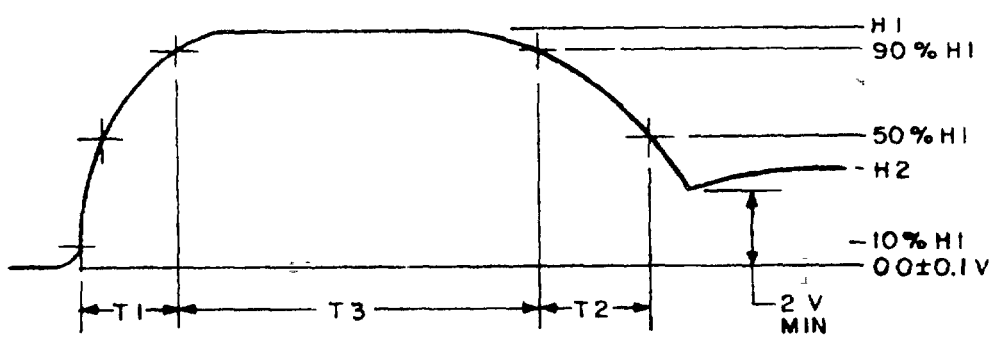
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NOTES

1. R1 = 178 Ω ±1%, 1/8 W
2. R2 = 301 Ω ±1%, 1/8 W
3. R3 = 649 Ω ±1%, 1/8 W
4. R4 = 154 Ω ±1%, 1/2 W
5. C1 = 22 µF ±10%
6. CR1 = 1N647 or equivalent
7. Relay-1 = GE 3SAV1256A2 or equivalent
8. Relay-2 = Agastat Model 2112 or equivalent
9. S1 = SPST switch
10. Resistors R1, R2, R3, and capacitor C1 are programming components for item PO1-2 and are based on a programming constant of 100 ohms per volt
11. Using item D03 with item Y04, adjust the time delay relay to obtain the output waveform (see figure 6) at pin 7 of the time delay relay after switch S1 is closed
12. For each test condition, switch S1 is closed and the voltage reading is made a minimum of 2 seconds after the closure of switch S1. At the conclusion of each test condition measurement, switch S1 is returned to the open position.
13. This circuit is drawn showing item PO1-2 with a negative reference terminal (non-commoned programming terminal labeled (-)). This circuit, as shown, will not operate properly if item PO1-2 has a positive non-commoned programming terminal. NOTE: Load current must not pass through the sensing network.

FIGURE 5. Zener applied potential test circuit for table III-Continued.

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NOTE Output waveform parameters
H1 = 11.25 V ±1%
H2 = 5.0 V ±10%
T1 = 100 +100, -50 ms
T2 = 100 +400, -50 ms
T3 = 0.5 s ±10%

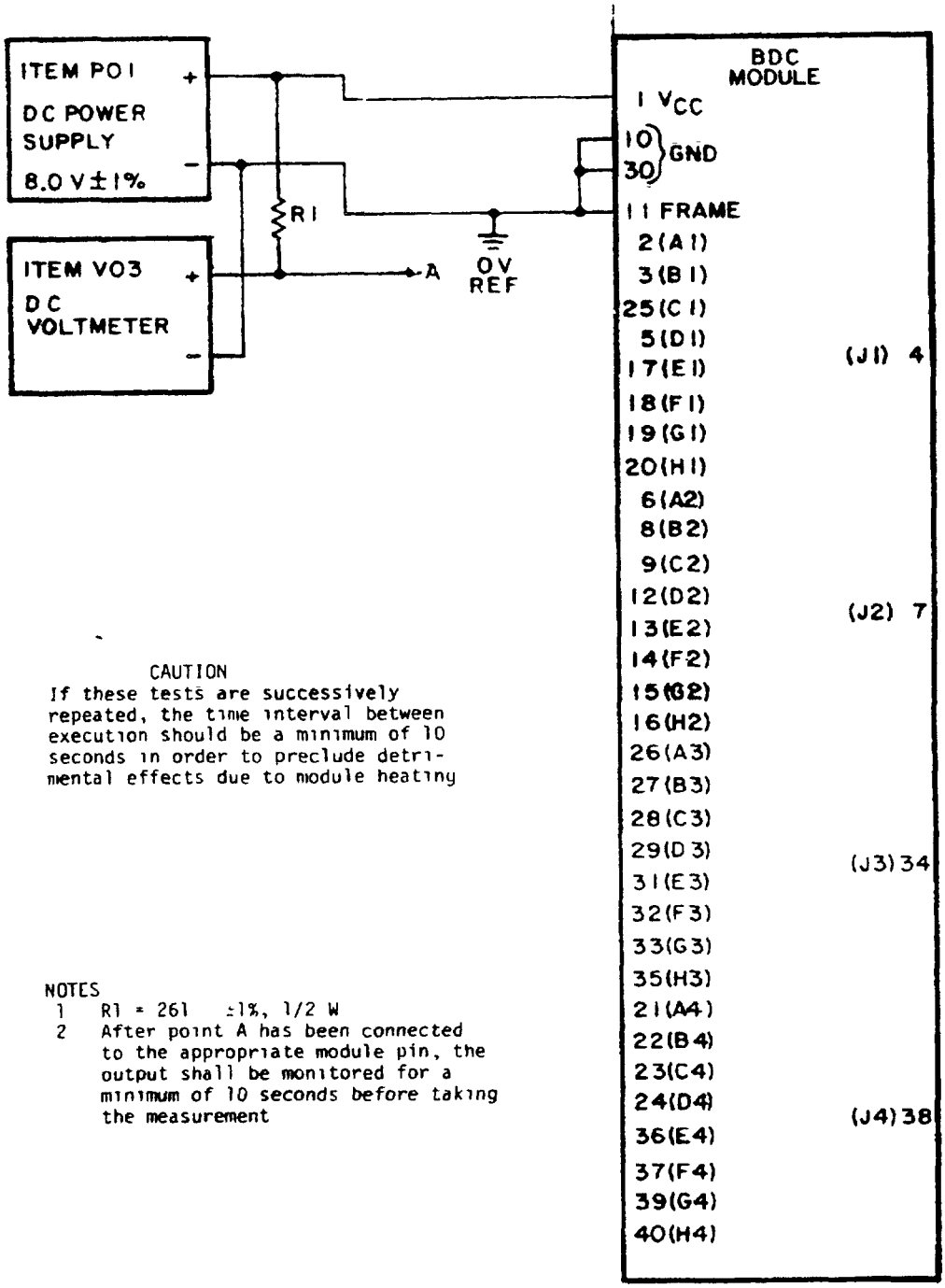
FIGURE 6 Zener applied potential test circuit waveform for table III

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TABLE IV V_{OUT} (low loaded) test

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		\ to pin	Min	Max		
C1	J1 V_{OUT} (low loaded)	4	0	1.1	V	7
C2	J2	7	0	1.1	V	7
C3	J3	34	0	1.1	V	7
C4	J4	38	0	1.1	V	7

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CAUTION
If these tests are successively repeated, the time interval between execution should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating

NOTES
1. R1 = 261 ±1%, 1/2 W
2. After point A has been connected to the appropriate module pin, the output shall be monitored for a minimum of 10 seconds before taking the measurement

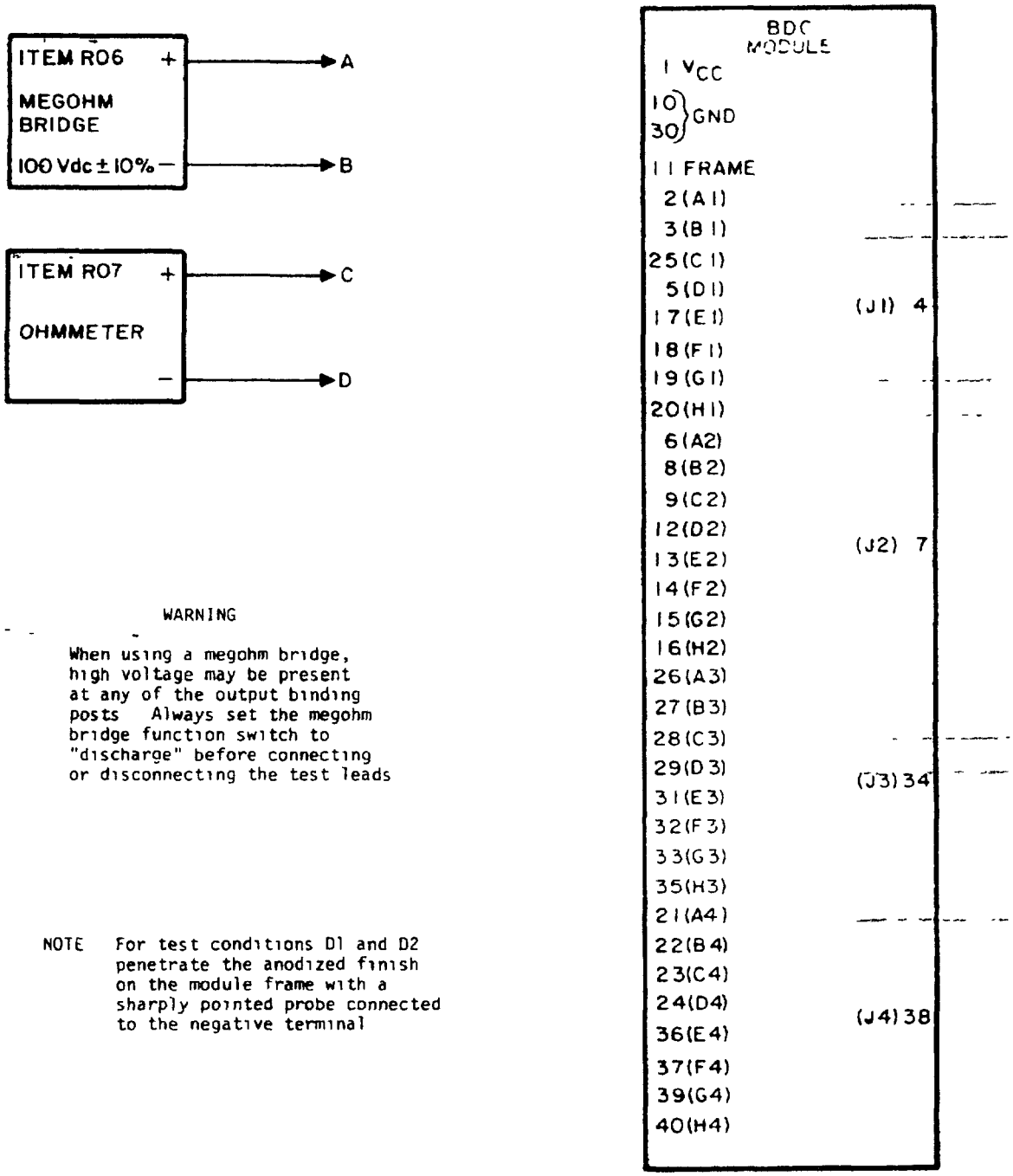
FIGURE 7. V_{OUT} (low loaded) test circuit for table IV

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TABLE V Isolation and continuity test

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Isolation	All module pins inter-connected	Frame	Open	Open	10	∞	MΩ	8
D2	Continuity	Open	Open	11	Frame	0.0	10	Ω	8

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test conditions D1 and D2 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 4. Isolation and continuity test circuit for table V.

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TABLE VI Output voltage (high-level and low-level) test

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figur
E1	J1 Output voltage, high level	2	Open	4	4	3 8	5 5	V	9
E2	J1	3							
E3	J1	25							
E4	J1	5							
E5	J1	17							
E6	J1	18							
E7	J1	19							
E8	J1	20							
E9	J2	6		7	7				
E10	J2	8							
E11	J2	9							
E12	J2	12							
E13	J2	13							
E14	J2	14							
E15	J2	15							
E16	J2	16							
E17	J3	26		34	34				
E18	J3	27		34	34				

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TABLE VI Output voltage (high-level and low-level) test - Continued

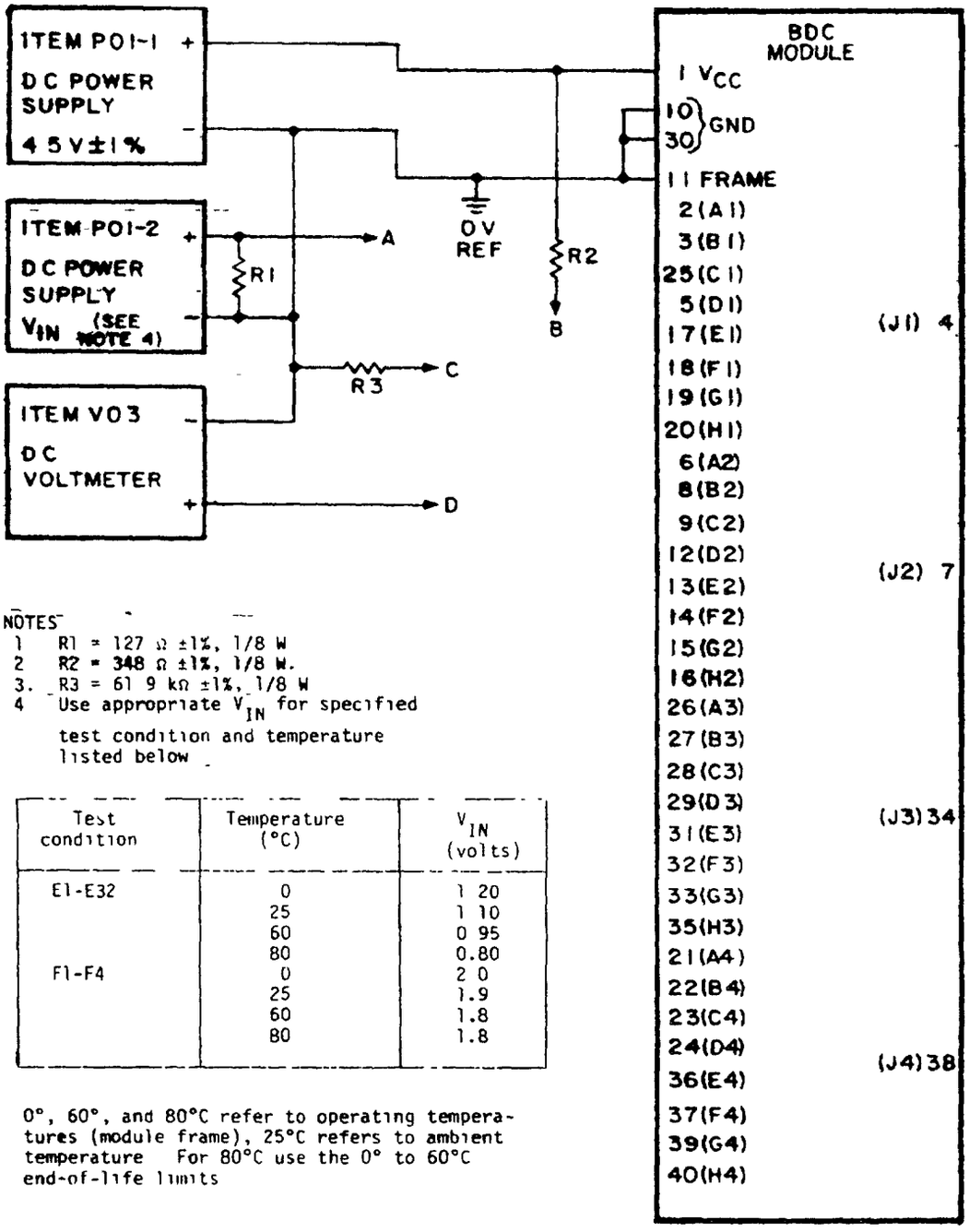
Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ±5°C ambient Min	Max	Unit	Test figure
E19	J3 Output voltage, high level	28	Open	34	34	3 8	5 5	V	9
E20	J3	29							
E21	J3	31							
E22	J3	32							
E23	J3	33							
E24	J3	35							
E25	J4	21		38	38				
E26	J4	22							
E27	J4	23							
E28	J4	24							
E29	J4	36							
E30	J4	37							
E31	J4	39							
E32	J4	40							
F1	J1 Output voltage, low level	2,3, 25,5, 17,18, 19,20	4	Open	4	0 00	0 40		
F2	J2 Output voltage, low level	8,6,9, 12,13, 14,15, 16	7	Open	7	0 00	0 40		

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TABLE VI Output voltage (high-level and low-level) test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F3	J3 Output voltage, low level	26,27, 28,29, 31,32, 33,35	34	Open	34	0 00	0 40	V	9
F4	J4 Output voltage, low level	21,22, 23,24, 36,37, 39,40	38	Open	38	0 00	0 40	V	9

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NOTES
1. R1 = 127 Ω ±1%, 1/8 W
2. R2 = 348 Ω ±1%, 1/8 W.
3. R3 = 61 9 kΩ ±1%, 1/8 W
4. Use appropriate V_{IN} for specified test condition and temperature listed below

Test condition	Temperature (°C)	V_{IN} (volts)
E1-E32	0	1 20
	25	1 10
	60	0 95
	80	0.80
F1-F4	0	2 0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module frame), 25°C refers to ambient temperature For 80°C use the 0° to 60°C end-of-life limits

FIGURE 9 Output voltage (high-level and low-level) test circuit for table VI

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TABLE VII Output short circuit current test

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	J1 Output short circuit current	Open	4	2,3,25,5, 17,18,19, 20	0 70	1 31	mA	10
G2	J2		7	6,8,9,12, 13,14,15, 16				
G3	J3		34	26,27,28, 29,31,32, 33,35				
G4	J4		38	21,22,23, 24,36,37, 39,40				

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TABLE VIII Input current (low-level and high-level) test

Test condition	Parameter	Connect			Initial limits 25° +5°C ambient		Unit	Test time
		A to pin	B to pin	C to pin	Min	Max		
H1	A1 Input current, low level	Open	2	Open	0.2	1.6	mA	10
H2	B1		3					
H3	C1		25					
H4	D1		5					
H5	E1		17					
H6	F1		18					
H7	G1		19					
H8	H1		20					
H9	A2		6					
H10	B2		8					
H11	C2		9					
H12	D2		12					
H13	E2		13					
H14	F2		14					
H15	G2		15					
H16	H2		16					
H17	A3		26					
H18	B3		27					
H19	C3		28					
H20	D3		29					

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connector			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H21	E3 Input current, low level	Open	31	Open	0.2	1.6	mA	10
H22	F3		32					
H23	G3		33					
H24	H3		35					
H25	A4		21					
H26	B4		22					
H27	C4		23					
H28	D4		24					
H29	E4		36					
H30	F4		37					
H31	G4		39					
H32	H4		40					
J1	A1 Input current, high level	2	Open	3	0	2	µA	
J2	A1			25				
J3	A1			5				
J4	A1			17				
J5	A1			18				
J6	A1			19				
J7	A1			20				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits -5° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J8	B1 Input current, high level	3	Open	2	0	2	µA	10
J9	B1			25				
J10	B1			5				
J11	B1			17				
J12	B1			18				
J13	B1			19				
J14	B1			20				
J15	C1	25		2				
J16	C1			3				
J17	C1			5				
J18	C1			17				
J19	C1			18				
J20	C1			19				
J21	C1			20				
J22	D1	5		2				
J23	D1			3				
J24	D1			25				
J25	D1			17				
J26	D1			18				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Conditions			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J27	D1 Input current, high level	5	Open	19	0	2	uA	10
J28	D1	5		20				
J29	E1	17		2				
J30	E1			3				
J31	E1			25				
J32	E1			5				
J33	E1			18				
J34	E1			19				
J35	E1			20				
J36	F1	18		2				
J37	F1			3				
J38	F1			25				
J39	F1			5				
J40	F1			17				
J41	F1			19				
J42	F1			20				
J43	G1	19		2				
J44	G1	19		3				
J45	G1	19		25				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test number	Parameter	Connect			Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J46	G1 Input current, high level	19	Open	5	0	2	μA	10
J47	G1			17				
J48	G1			18				
J49	G1			20				
J50	H1	20		2				
J51	H1			3				
J52	H1			25				
J53	H1			5				
J54	H1			17				
J55	H1			18				
J56	H1			19				
J57	A2	6		8				
J58	A2			9				
J59	A2			12				
J60	A2			13				
J61	A2			14				
J62	A2			15				
J63	A2			16				
J64	B2	8		6				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J65	B2 Input current, high level	8	Open	9	0	2	µA	10
J66	B2			12				
J67	B2			13				
J68	B2			14				
J69	B2			15				
J70	B2			16				
J71	C2	9		6				
J72	C2			8				
J73	C2			12				
J74	C2			13				
J75	C2			14				
J76	C2			15				
J77	C2			16				
J78	D2	12		6				
J79	D2			8				
J80	D2			9				
J81	D2			13				
J82	D2			14				
J83	D2			15				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test section	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J84	D2 Input current, high level	12	Open	16	0	2	µA	10
J85	E2	13		6				
J86	E2			8				
J87	E2			9				
J88	E2			12				
J89	E2			14				
J90	E2			15				
J91	E2			16				
J92	F2	14		6				
J93	F2			8				
J94	F2			9				
J95	F2			12				
J96	F2			13				
J97	E2			15				
J98	F2			16				
J99	G2	15		6				
J100	G2			8				
J101	G2			9				
J102	G2			12				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
J103	G2 Input current, high level	15	Open	13	0	2	µA	10
J104	G2	15		14				
J105	G2	15		16				
J106	H2	16		6				
J107	H2			8				
J108	H2			9				
J109	H2			12				
J110	H2			13				
J111	H2			14				
J112	H2			15				
J113	A3	26		27				
J114	A3			28				
J115	A3			29				
J116	A3			31				
J117	A3			32				
J118	A3			33				
J119	A3			35				
J120	B3	27		26				
J121	B3	27		28				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits -55°C to +125°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J122	B3 Input current, high level	27	Open	29	0	2	µA	10
J123	B3			31				
J124	B3			32				
J125	B3			33				
J126	B3			35				
J127	C3	28		25				
J128	C3			27				
J129	C3			29				
J130	C3			31				
J131	C3			32				
J132	C3			33				
J133	C3			35				
J134	D3	29		26				
J135	D3			27				
J136	D3			28				
J137	D3			31				
J138	D3			32				
J139	D3			33				
J140	D3			35				

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TABLE VIII Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J141	E3 Input current, high level	31	Open	26	0	2	µA	10
J142	E3			27				
J143	E3			28				
J144	E3			29				
J145	E3			32				
J146	E3			33				
J147	E3			35				
J148	F3	32		26				
J149	F3			27				
J150	F3			28				
J151	F3			29				
J152	F3			31				
J153	F3			33				
J154	F3			35				
J155	G3	33		26				
J156	G3			27				
J157	G3			28				
J158	G3			29				
J159	G3			31				

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TABLE VIII. Input current (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J160	G3 Input current, high level	33	Open	32	0	2	µA	10
J161	G3	33		35				
J162	H3	35		26				
J163	H3			27				
J164	H3			28				
J165	H3			29				
J166	H3			31				
J167	H3			32				
J168	H3			33				
J169	A4	21		22				
J170	A4			23				
J171	A4			24				
J172	A4			36				
J173	A4			37				
J174	A4			39				
J175	A4			40				
J176	B4	22		21				
J177	B4	22		23				
J178	B4	22		24				

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TABLE VIII Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J179	B4 Input current, High level	22	Open	36	0	2	µA	10
J180	B4			37				
J181	B4			39				
J182	B4			40				
J183	C4	23		21				
J184	C4			22				
J185	C4			24				
J186	C4			36				
J187	C4			37				
J188	C4			39				
J189	C4			40				
J190	D4	24		21				
J191	D4			22				
J192	D4			23				
J193	D4			36				
J194	D4			37				
J195	D4			39				
J196	D4			40				
J197	E4	36		21				

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TABLE VIII Input current (low-level and high-level) test - Continued

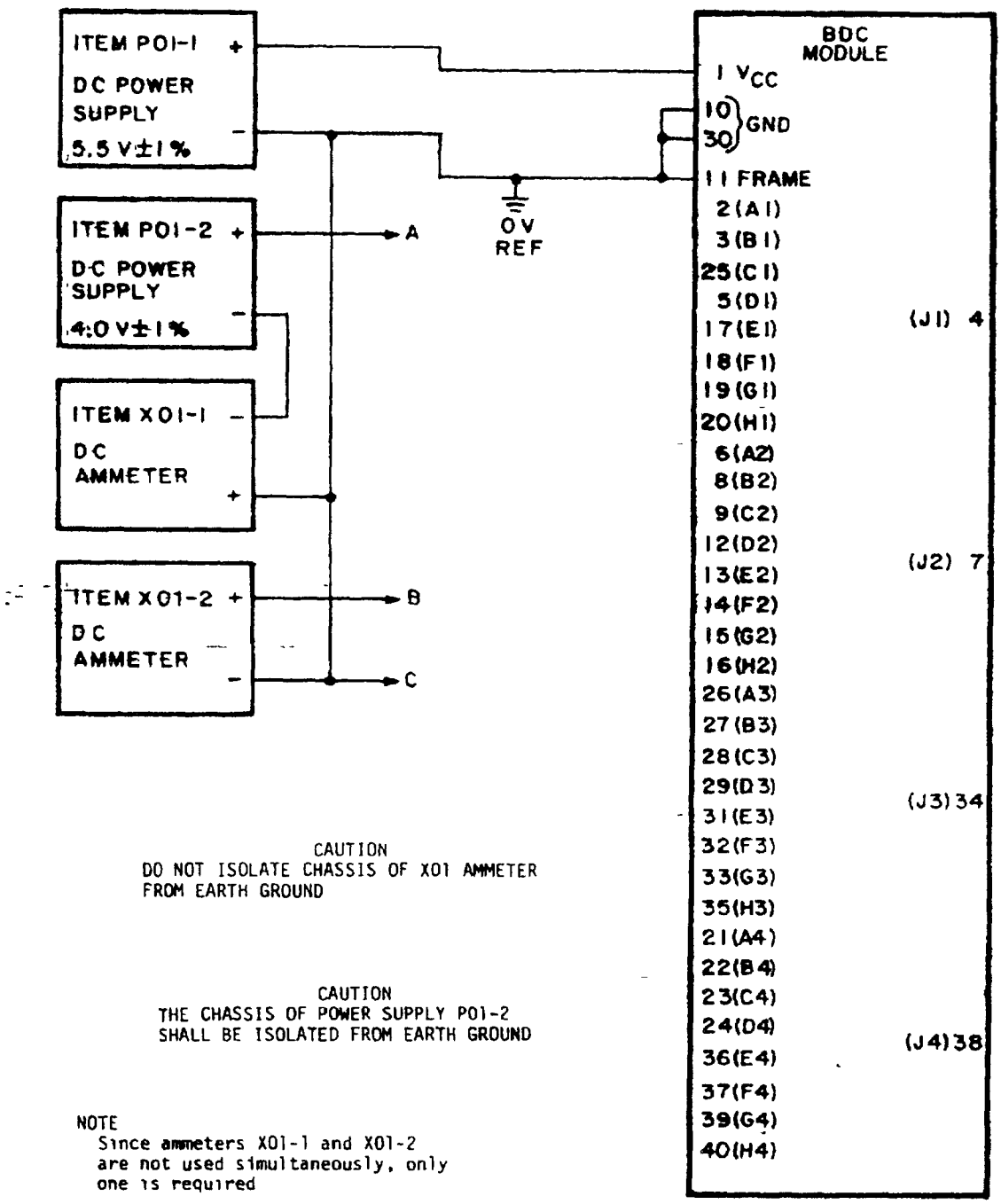
Set condition	Parameter	Connect			Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J198	E4 Input current, high level	36	Open	22	0	2	µA	10
J199	E4			23				
J200	E4			24				
J201	E4			37				
J202	E4			39				
J203	E4			40				
J204	F4	37		21				
J205	F4			22				
J206	F4			23				
J207	F4			24				
J208	F4			36				
J209	F4			39				
J210	F4			40				
J211	G4	39		21				
J212	G4			22				
J213	G4			23				
J214	G4			24				
J215	G4			36				
J216	G4			37				
J217	G4			40				

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TABLE VIII Input current (low-level and high-level) test - Continued.

Test conditions	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J218	H4 Input current, high level	40	Open	21	0	2	µA	10
J219	H4			22				
J220	H4			23				
J221	H4			24				
J222	H4			36				
J223	H4			37				
J224	H4			39				

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CAUTION
DO NOT ISOLATE CHASSIS OF X01 AMMETER FROM EARTH GROUND

CAUTION
THE CHASSIS OF POWER SUPPLY P01-2 SHALL BE ISOLATED FROM EARTH GROUND

NOTE
Since ammeters X01-1 and X01-2 are not used simultaneously, only one is required

FIGURE 10. Output short circuit current and input current (low-level and high-level) test circuit for Tables VII and VIII

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TABLE IX. Input capacitance test

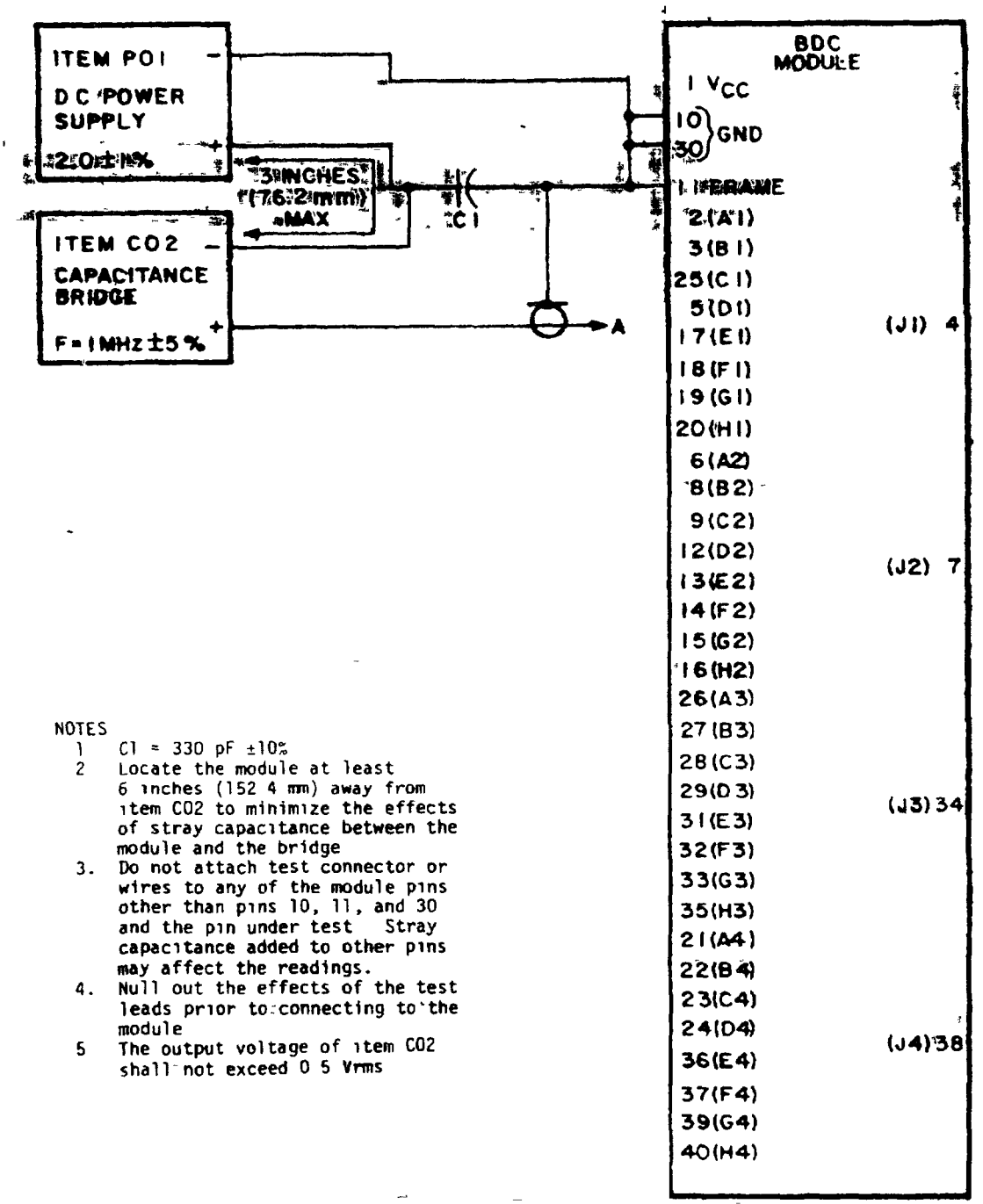
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	A1 Input capacitance	2	2	16	pF	11
K2	B1	3				
K3	C1	25				
K4	D1	5				
K5	E1	17				
K6	F1	18				
K7	G1	19				
K8	H1	20				
K9	A2	6				
K10	B2	8				
K11	C2	9				
K12	D2	12				
K13	E2	13				
K14	F2	14				
K15	G2	15				
K16	H2	16				
K17	A3	26				
K18	B3	27				

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TABLE IX Input capacitance test - Continued

Test condition	Parameter	Connect	Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K19	C3 Input capacitance	28	2	16	pF	11
K20	D3	29				
K21	E3	31				
K22	F3	32				
K23	G3	33				
K24	H3	35				
K25	A4	21				
K26	B4	22				
K27	C4	23				
K28	D4	24				
K29	E4	36				
K30	F4	37				
K31	G4	39				
K32	H4	40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:06Z
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NOTES
1. C1 = 330 pF ±10%
2. Locate the module at least 6 inches (152.4 mm) away from item CO2 to minimize the effects of stray capacitance between the module and the bridge
3. Do not attach test connector or wires to any of the module pins other than pins 10, 11, and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
4. Null out the effects of the test leads prior to connecting to the module
5. The output voltage of item CO2 shall not exceed 0.5 Vrms

FIGURE 11. Input capacitance test circuit for table IX

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TABLE X. Power supply current (high-level and low-level) test

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, high level	10	0	11	mA	12
L2	Power supply current, low level	1	0 0	17 6	mA	12

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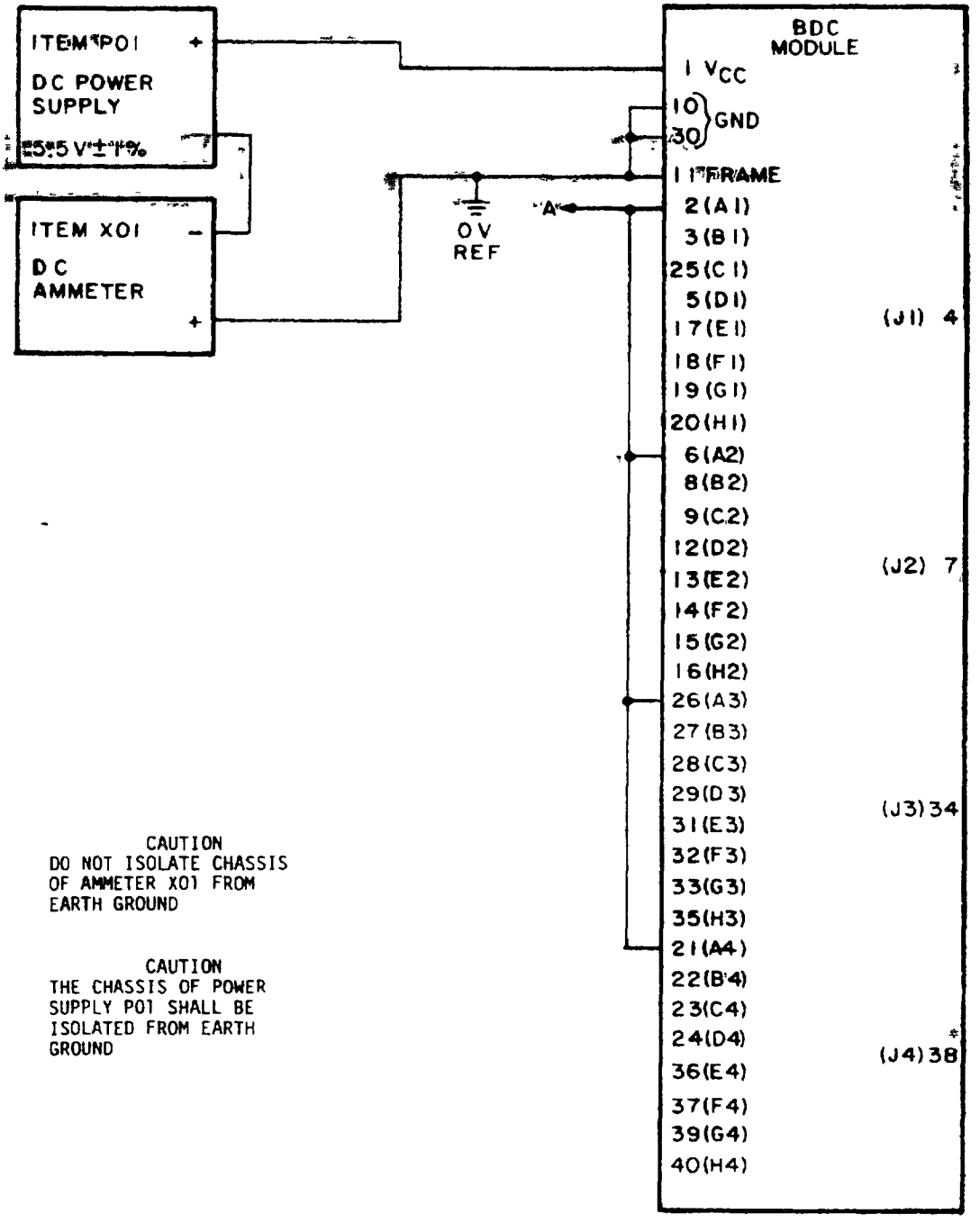


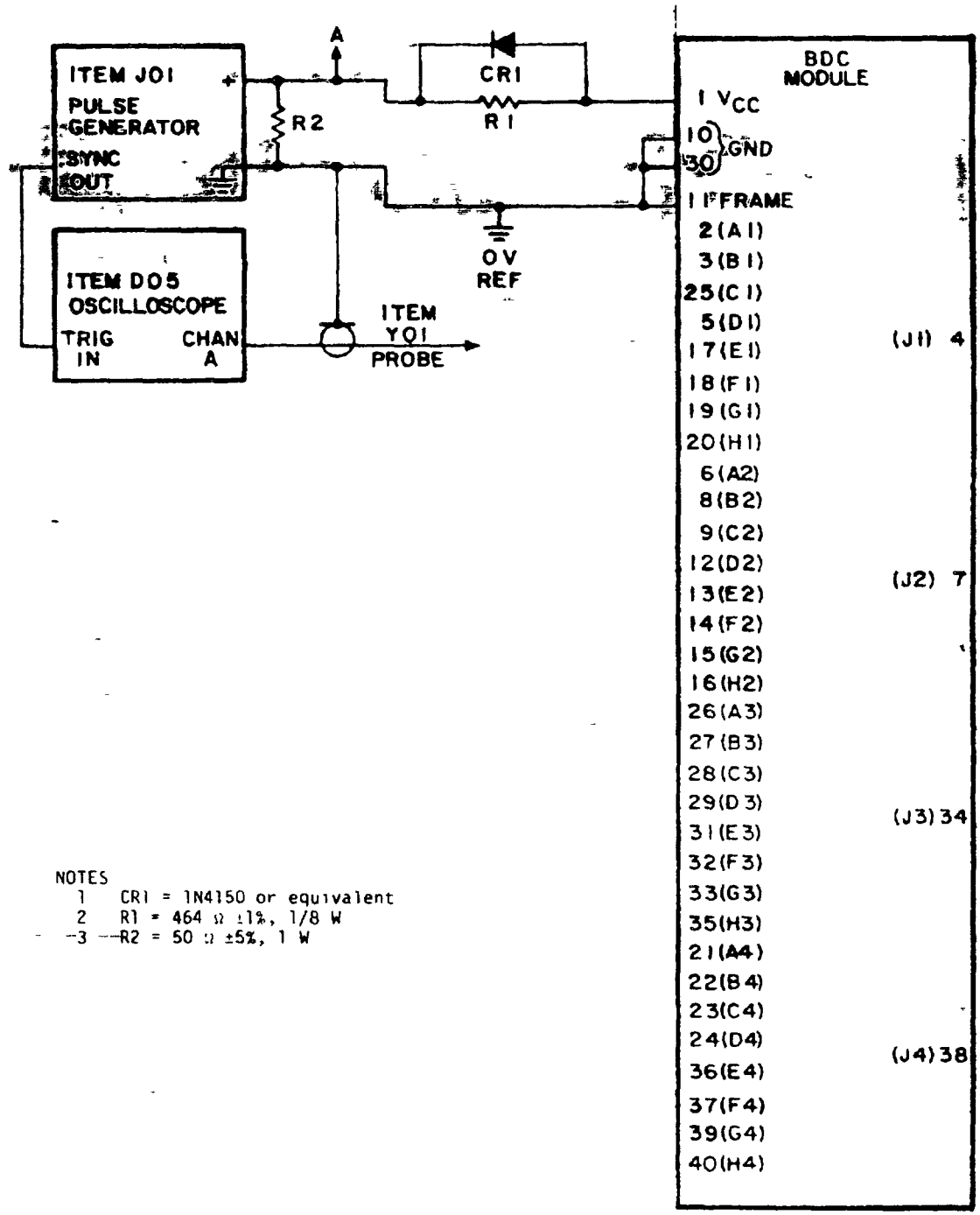
FIGURE 12 Power supply current (high-level and low-level) test circuit for table X

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TABLE XI Power supply transient rise time test

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply transient rise time	36	148	us	13, 14

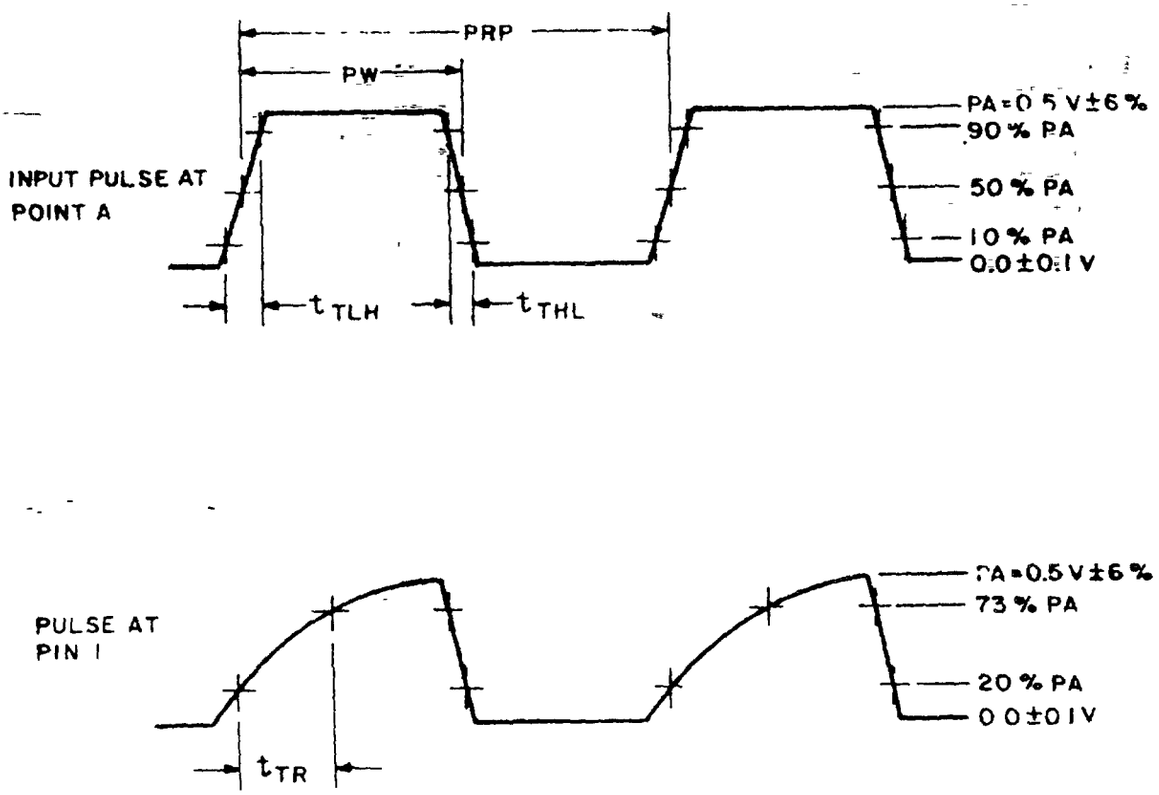
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:06Z
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NOTES
1. CR1 = 1N4150 or equivalent
2. R1 = 464 Ω ±1%, 1/8 W
3. R2 = 50 Ω ±5%, 1 W

FIGURE 13. Power supply transient rise time test circuit for table XI

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NOTE Input pulse parameters

PW = 750 us ±5% (at 50% amplitude)
PRP = 1.5 ms ±5%
$t_{TLH} = t_{THL}$ = 50 ns ±10%.

FIGURE 14 Power supply transient rise time test circuit waveforms for table XI

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TABLE XII Output pulse amplitude (high-level and low-level) test

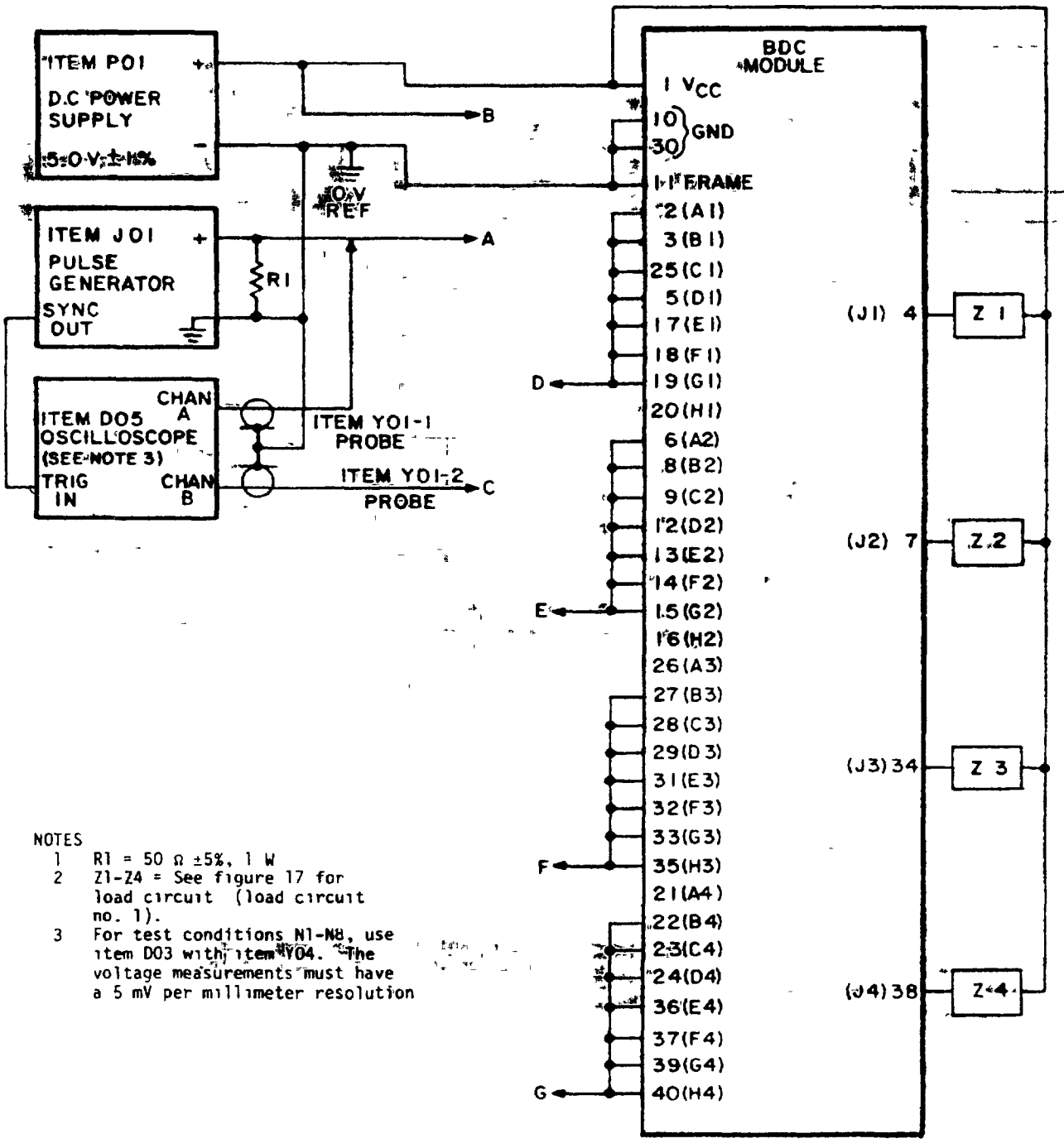
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Notes
		A to pin	B to pin	C to pin	Min	Max		
N1	J1 Output pulse amplitude, high level	20	D	4	3.8	5.5	V	15, 16
N2	J2	16	E	7				
N3	J3	26	F	34				
N4	J4	21	G	38				
N5	J1 Output pulse amplitude, low level	20	D	4	0.00	0.40		
N6	J2	16	E	7				
N7	J3	26	F	34				
N8	J4	21	G	38				

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TABLE XIII. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test

Test condition	Parameter	Connect			Initial limits -55° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	J1 Transition time, (t_{THL})	20	D	4	6	24	ns	15, 16
P2	J2	16	E	7				
P3	J3	26	F	34				
P4	J4	21	G	38				
P5	J1 Transition time, (t_{TLH})	20	D	4	60	160		
P6	J2	16	E	7				
P7	J3	26	F	34				
P8	J4	21	G	38				
P9	J1 Propagation delay time (t_{PLH})	20	D	4	20	75		
P10	J2	16	E	7				
P11	J3	26	F	34				
P12	J4	21	G	38				
P13	J1 Propagation delay time (t_{PHL})	20	D	4	6	70		
P14	J2	16	E	7				
P15	J3	26	F	34				
P16	J4	21	G	38				

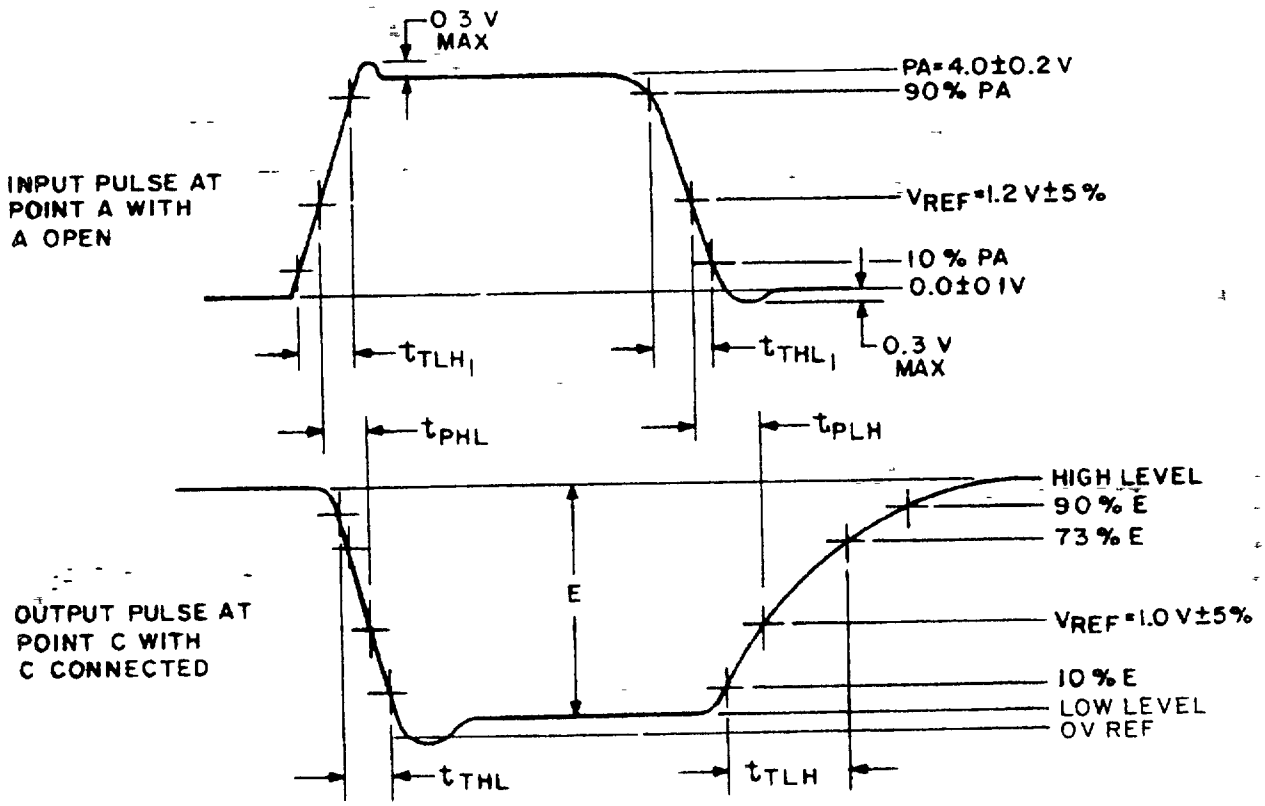
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NOTES
1. R1 = 50 Ω ±5%, 1 W
2. Z1-Z4 = See figure 17 for load circuit (load circuit no. 1).
3. For test conditions N1-N8, use item D03 with item Y04. The voltage measurements must have a 5 mV per millimeter resolution

FIGURE 15 Output pulse amplitude (high-level and low-level), transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII and XIII

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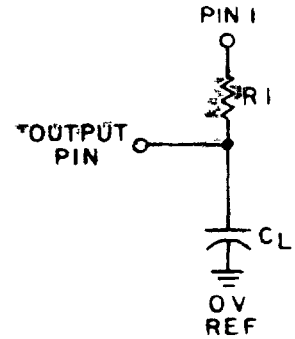


NOTE Input pulse parameters

PW = 1 µs ±10%
$t_{TLH_1} = t_{THL_1}$ = 20 ±2 ns
PRR = 100 kHz ±10%

FIGURE 16 Output pulse amplitude (high-level and low-level), transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII and XIII

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LOAD NO 1

NOTES

1 For test conditions N1-N8 and P1-P16

(a) R1 = 3.83 kΩ ±1%, 1/8 W

(b) C_L = 30 ±5 pF (includes stray, probe, and compensating capacitor, as required).

(c) With the module unplugged, the oscilloscope probe disconnected, and the 3.83 kΩ load resistors disconnected from their respective output pins, use item C02 to measure the capacitance between each test connector output pin and pins 10, 11, and 30 combined. The resistors shall be disconnected from their respective output pins as close to their bodies as practical. The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin. The capacitance shall be measured at 1 MHz ±10%. Reconnect the 3.83 kΩ resistors prior to using the test connector for module testing.

(d) The loop length from pin 1 to each load resistor and output, including the length of the resistor body, shall not exceed 4.5 inches (114.3 mm).

LOAD NO 2

2 For test conditions Q1-Q4 and R1-R4

(a) R1 = 422 Ω ±1%, 1/8 W

(b) Same as 1(b)

(c) Same as 1(c)

(d) Same as 1(d)

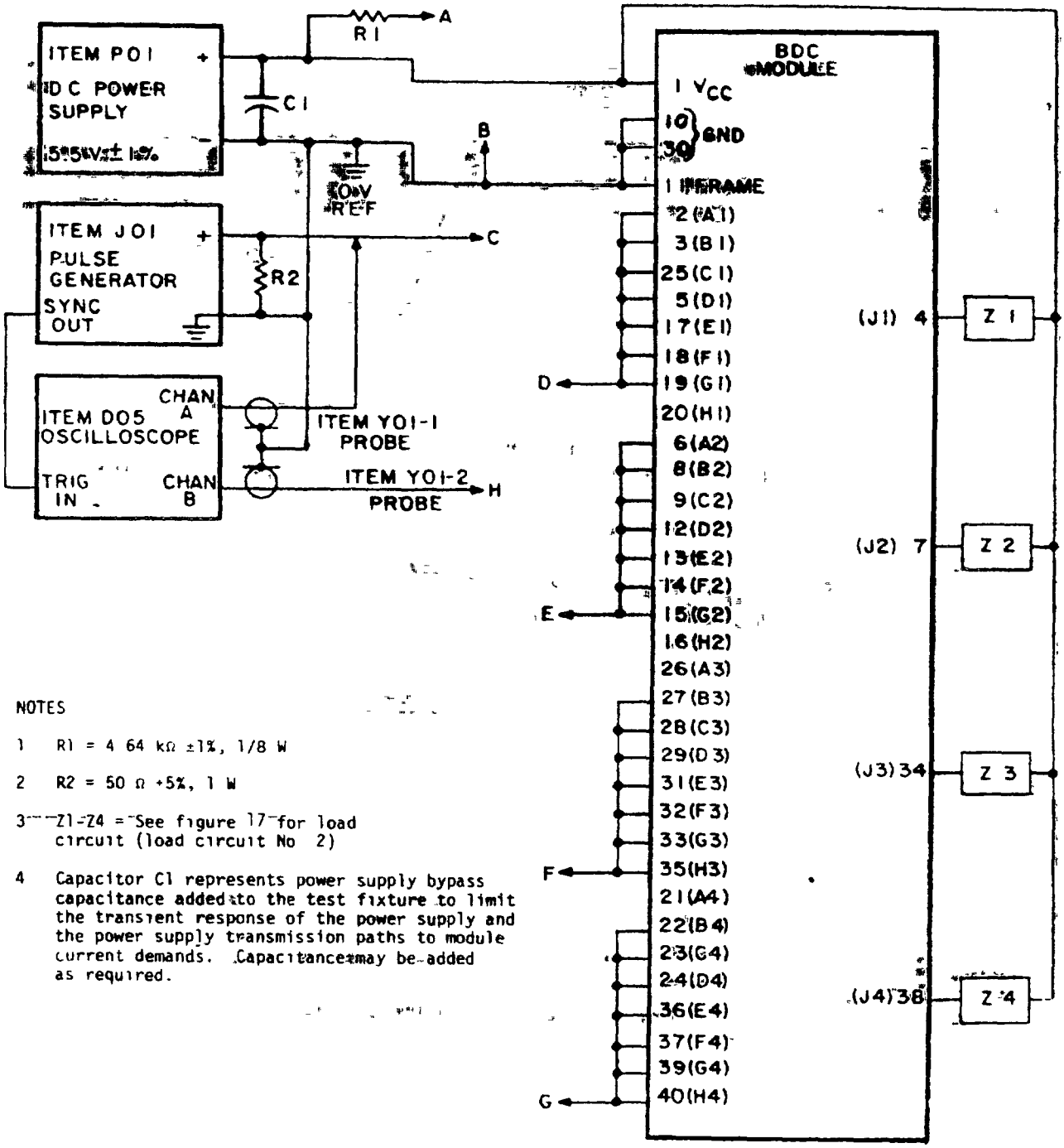
FIGURE 17 BDC logic load circuit

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TABLE XIV Output crosstalk (high-level and low-level) test

Test condition	Parameter	Connect'i				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	J1 Output crosstalk, high level	Open	D	E,F,G	4	0 0	0 8	V	18, 19
Q2	J2		E	D,F,G	7				
Q3	J3		F	D,E,G	34				
Q4	J4		G	D,E,F	38				
R1	J1 Output crosstalk, low level	0	Open	E,F,G	4				
R2	J2	E		D,F,G	7				
R3	J3	F		D,E,G	34				
R4	J4	G		D,E,F	38				

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NOTES

1 R1 = 4 64 kΩ ±1%, 1/8 W

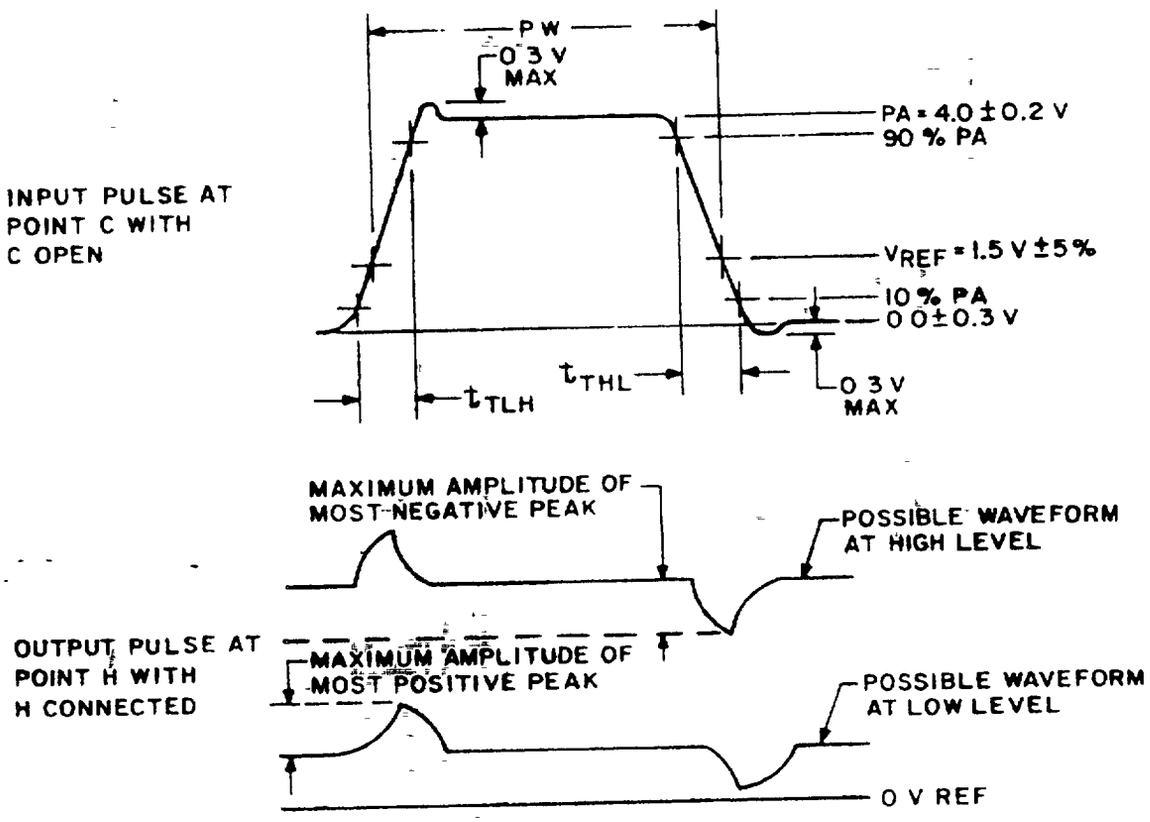
2 R2 = 50 Ω ±5%, 1 W

3 Z1-Z4 = See figure 17 for load circuit (load circuit No 2)

4 Capacitor C1 represents power supply bypass capacitance added to the test fixture to limit the transient response of the power supply and the power supply transmission paths to module current demands. Capacitance may be added as required.

FIGURE 18 Output crosstalk (high-level and low-level) test circuit for table XIV

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NOTE Input pulse parameters

PW = 0.9 ±0.1 µs
$t_{TLH} = t_{THL}$ = 20 ±2 ns
PRR = 100 kHz ±10%

FIGURE 19 Output crosstalk (high-level and low-level) test circuit waveforms for table XIV

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